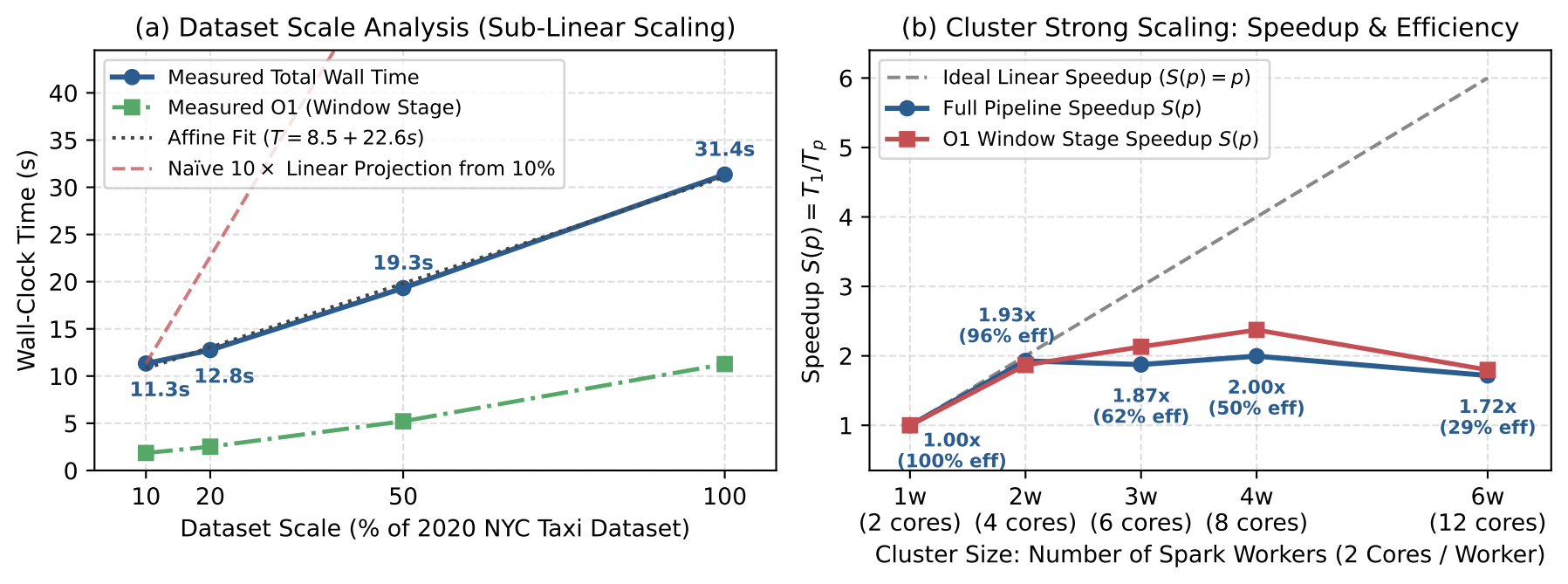
(a) Dataset Scale Analysis (Sub-Linear Scaling)
Measured Total Wall Time
Measured O1 (Window Stage)
Affine Fit ($T = 8.5 + 22.6s$)
Naïve 10 × Linear Projection from 10%
Wall-Clock Time (s)
Dataset Scale (% of 2020 NYC Taxi Dataset)
11.3s
12.8s
19.3s
31.4s
(b) Cluster Strong Scaling: Speedup & Efficiency
Ideal Linear Speedup ($S(p) = p$)
Full Pipeline Speedup $S(p)$
O1 Window Stage Speedup $S(p)$
Speedup $S(p) = T_1/T_p$
1.00x (100% eff)
1.93x (96% eff)
1.87x (62% eff)
2.00x (50% eff)
1.72x (29% eff)
1w (2 cores)
2w (4 cores)
3w (6 cores)
4w (8 cores)
6w (12 cores)
Cluster Size: Number of Spark Workers (2 Cores / Worker)